

On the consistency of kernel density estimates under modality constraints

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Abstract

Nonparametric density estimation under modality constraints has attracted considerable interest. With classical kernel density estimates, the number of modes depends on the chosen bandwidth. We consider the Gaussian kernel and prove for almost arbitrary and possibly non-smooth densities that there is a sequence of ranges of bandwidths leading to consistent estimates while still guaranteeing either the correct number of modes k or a correct upper bound on k . More precisely, a bandwidth sequence that is selected by any method from our proposed sequence of ranges will lead to estimates that are consistent at continuity points.

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1. Introduction

When knowledge concerning qualitative properties of probability densities is available, it is often desirable that density estimates reflect these properties. Indeed, from applied point of view, ensuring the correct representation of qualitative properties can often be more important than the minimization of some quantitative performance measure like the MSE.

A frequent qualitative assumption is unimodality or, more generally, k -modality. Applications involving densities with a known number of modes (or at least a known upper bound on the number of modes) occur for instance in the context of mixture models when there is some knowledge concerning the maximum possible number of mixture components. As another example, consider intensity densities of non-homogeneous Poisson processes, where there is often some qualitative knowledge concerning the phenomenon investigated (think for instance of a known periodicity) suggesting a certain number of inflection points. Notice that conditional on the total number of observations, the jump points of a non-homogeneous Poisson process behave like i.i.d. observation taken from its intensity density.

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From a practical point of view, the probably easiest and most straightforward approach to obtain a density with the desired number of modes is to use a kernel density estimate

$$\hat{f}_{n,h}(x) := \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right), \quad (1)$$

based on n i.i.d. observations $X_1, \dots, X_n$, and with an appropriately chosen bandwidth. Indeed, the number of modes depends crucially on the bandwidth and seems to be one of the most important criteria when a bandwidth is chosen “by eye”. While this is true in general, it has been shown in particular for the Gaussian kernel that the number of modes is a decreasing and right continuous function of the bandwidth (see [Silverman, 1981](#)). Thus the set of bandwidths leading to k modes is some half-open interval $S_n = [H_{n,k}, H_{n,k-1})$. For any k , the bound $H_{n,k}$ may be interpreted as the smallest bandwidth leading to k modes. If k is only some known upper bound to the number of modes, any bandwidth chosen from $S_n^{(U)} = [H_{n,k}, \infty)$ can enforce this bound.

It should be noted that the sets S_n and $S_n^{(U)}$ will not necessarily include bandwidths that optimize quantitative criteria like the MISE. Indeed, think for instance of densities that are very flat in some regions, but have substantial curvature in others. Then ensuring an appropriate number of modes usually requires substantial smoothing where f is flat, causing considerable bias in the curved regions. [Hart \(1984\)](#) made a similar observation when looking at the number of modes exhibited by the expected value of a density estimate. On the other hand, imposing modality constraints on data driven methods of bandwidth choice often leads to good results (see [Futschik and Clarke, 2004](#)). Overall, we view the approach enforcing modality constraints by a suitable bandwidth choice as a quick and—given existing statistical software—easy to use exploratory tool that frequently leads to reasonable results, while satisfying some minimal quality requirements under very general conditions.

There are several often more sophisticated alternative approaches to estimate a density under modality constraints. Modifications of kernel density estimates for smooth unimodal densities have been proposed both for the case of a known mode (see [Fougères, 1997](#)) and when the mode is unknown ([Eggermont and LaRiccia, 2000](#); [Hall and Huang, 2002](#)). Further related work includes smoothing under assumptions on the number of inflection points in the regression context (see [Mammen, 1991](#); [Riedel, 1997](#)) and hypotheses testing concerning the number of modes (see [Silverman, 1981](#); [Mammen, 1995](#)). For non-smooth unimodal densities, a piecewise application of the Grenander estimate for monotone densities is another alternative. Estimates obtained in this way may be viewed as a combination of the nonparametric maximum likelihood approach with criteria for selecting the mode. Proposals by [Wegman \(1969, 1970a,b\)](#), [Bickel and Fan \(1996\)](#) as well as [Birgé \(1997\)](#) may be viewed as examples of this approach. Furthermore [Cheng et al. \(1999\)](#) proposed a recursive estimate for unimodal densities.

The goal of this paper is to show that under minimal conditions, consistent estimates can be obtained while ensuring the correct number of modes, or a correct upper bound on the actual number of modes. Indeed, we propose subsets of S_n or $S_n^{(U)}$ such that any (subjective or objective) bandwidth choice taken from these sets will lead to estimates that are consistent at continuity points.

We will only require that the unknown density f is bounded and strictly k -modal, which is that f has k local maxima and $k - 1$ local minima and is strictly increasing (resp. decreasing) between adjacent local extrema. We do not require that f is differentiable, or even continuous and also do not make any requirements concerning the tails or the flatness of f . Notice, however, that piecewise monotonous functions are almost everywhere continuous and even differentiable (see for instance [Billingsley, 1986](#), Theorem 31.2.).

Under conditions permitting the uniform approximation of f and its first two derivatives, and under further conditions such as a non-vanishing second derivative of f at extremal points, [Mammen et al. \(1992\)](#) provide asymptotic approximations to $H_{n,k}$. In their setting, the lower bound $H_{n,k}$ of S_n and $S_n^{(U)}$ has the order of magnitude $n^{-1/5}$. This is also the order of magnitude of the MISE optimal bandwidth assuming two continuous derivatives of f , indicating that the correct number of modes can be achieved while still using a bandwidth of optimal order. However, their method of proof heavily relies on the uniform approximation of f and its derivatives by $\hat{f}_{n,h}$ and does not work with non-smooth densities.

Finally, for piecewise convex–concave densities, Cuevas and Gonzalez-Manteiga (1991) proposed to use kernel estimates with bandwidths chosen to obtain equality of the estimate and the true density with respect to the number of inflection points. They consider in particular estimates based on the Gaussian kernel and the smallest possible bandwidth $H_{n,k}$ that provides the desired number of inflection points. For their density estimate, they prove strong uniform consistency but have to assume the existence of a bandwidth sequence $h_n \rightarrow 0$ satisfying their modality conditions. Our Lemma 1 in Section 2 implies that $H_{n,k}$ satisfies their requirement almost surely.

2. The range of bandwidths satisfying a certain modality constraint

Our results are for the kernel density estimate $\hat{f}_{n,h}$ defined in (1) when $K(x) = (1/\sqrt{2\pi})e^{-x^2/2}$ is chosen to be the Gaussian kernel. We make the following weak assumptions.

- (A1) f has k_f local maxima at some unknown points $\theta_1, \dots, \theta_{k_f}$ and $k_f - 1$ local minima at unknown points $\xi_1, \dots, \xi_{k_f-1}$;
- (A2) f is nowhere constant;
- (A3) f is bounded.

These assumptions imply that f is strictly increasing on the intervals $[\xi_i, \theta_{i+1}]$ for $1 \leq i \leq k_f - 1$ and strictly decreasing on the intervals $[\theta_i, \xi_i]$ ($1 \leq i \leq k_f - 1$). Furthermore, f is strictly increasing on $(-\infty, \theta_1]$ and strictly decreasing on $[\theta_{k_f}, \infty)$. We do not make any smoothness assumptions.

Recall that $H_{n,k}$ denotes the smallest bandwidth h such that the resulting estimate $\hat{f}_{n,h}$ has at most k modes. The next result establishes that $H_{n,k} \rightarrow 0$ a.s. under (A1)–(A3). Since the proof of the result is rather long, we give a brief outline here. Due to lack of smoothness our proof cannot be based on uniform approximation results for f and its derivatives. We therefore focus on

$$g_h(x) := \mathbb{E} \hat{f}_{n,h}(x) = \frac{1}{h} \int K\left(\frac{x-u}{h}\right) f(u) du, \quad (2)$$

which can be easily seen to have derivatives of any order. The main part of the proof consists in establishing that for some arbitrarily small h , the conditions

- (P1) g'_h changes sign at exactly $2k_f - 1$ points $\gamma_1^{(h)}, \dots, \gamma_{2k_f-1}^{(h)}$;
- (P2) if $g'_h(x) = 0$ then $x \in A := \{\gamma_1^{(h)}, \dots, \gamma_{2k_f-1}^{(h)}\}$;
- (P3) $g''_h(x) \neq 0$ for $x \in A$

are satisfied. After proving these properties, the result follows by establishing uniform convergence of $\hat{f}_{n,h}$ to g_h for some arbitrarily small fixed h using arguments analogous to those in Mammen et al. (1992).

Lemma 1. For $k \geq k_f$, we have that $H_{n,k} \rightarrow 0$ a.s., under the assumptions (A1)–(A3).

Proof. We start by establishing uniform convergence of $\hat{f}_{n,h}^{(v)}$ to $g_h^{(v)}$ for $v = 0, 1, 2$, i.e. up to the second derivative. Notice for this purpose that with $V(\cdot)$ denoting the total variation and for arbitrary constant h

$$\begin{aligned} & \sup_x |\hat{f}_{n,h}^{(v)}(x) - g_h^{(v)}(x)| \\ &= \sup_x \left| \frac{1}{h^{v+1}} \int_{-\infty}^{\infty} K^{(v)}\left(\frac{x-u}{h}\right) d[F_n(u) - F(u)] \right| \\ &\leq \frac{1}{h^{v+1}} \|F_n - F\|_{\infty} V\left[K^{(v)}\left(\frac{x-u}{h}\right)\right], \end{aligned} \quad (3)$$

which tends to zero almost surely by the Glivenko–Cantelli theorem. Thus

$$\sup_x |\hat{f}_{n,h}^{(v)}(x) - g_h^{(v)}(x)| \rightarrow 0 \quad \text{a.s.}, \quad (4)$$

for the derivatives $v = 0, 1, 2$.

Choose now an otherwise arbitrary $\varepsilon > 0$ that is small enough such that g_h has at least k_f local maxima for all $h < \varepsilon$. This is possible since $g_h(x) \rightarrow f(x)$ pointwise at continuity points as $h \rightarrow 0$. Thus, since piecewise monotonous functions are continuous except for a set of measure zero, there must be a local extremum of g_h in some arbitrarily small neighborhood of each local maximum θ_i ($1 \leq i \leq k_f$) of f , given that h is small enough. We will see later that the number of sign changes of g_h is exactly k_f for any $h < \varepsilon$.

Suppose furthermore that a fixed $h < \varepsilon$ is chosen such that (P1)–(P3) hold. Assertion (4) then implies for this fixed h that $\hat{f}_{n,h}'$ has $2k_f - 1$ sign changes a.s. for large enough n . Thus $0 \leq \limsup_n H_{n,k} \leq \limsup_n H_{n,k_f} \leq h < \varepsilon$ a.s., implying $H_{n,k} \rightarrow 0$ a.s.

It remains to show (P1)–(P3) for some $h < \varepsilon$.

To prove (P1), we approximate f by a sequence of functions $(f_m)_{m \geq 2}$ that are easier to handle than f . Take for instance $a_m = -\sqrt{m}$, $b_m = \sqrt{m}$, and

$$\mathcal{G} = \{x_i := a_m + (b_m - a_m)i/m; 0 \leq i \leq m\}.$$

Define $f_m(x_i) = f(x_i)$ for $x_i \in \mathcal{G}$ and use linear interpolation in between. Outside $[x_0, x_m]$, define $f_m(x) = f(x_m)(x_m/x)$ if $x > x_m$, and $f_m(x) = f(x_0)(x_0/x)$ if $x < x_0 < 0$. If the resulting f_m is constant on some subinterval then add $(x_j + x_{j+1})/2$ to $\mathcal{G}$ and define f_m by interpolation based on the augmented grid. Repeat this until the function f_m obtained this way is nowhere constant. Define furthermore

$$g_{m,h}(x) = \frac{1}{h} \int K\left(\frac{x-u}{h}\right) f_m(u) du.$$

Then $f_m \rightarrow f$ a.e. and, for any fixed x and h , and for $m \rightarrow \infty$

$$\begin{aligned} g_{m,h}'(x) &= \frac{1}{h^2} \int K'\left(\frac{x-u}{h}\right) f_m(u) du \\ &\rightarrow \frac{1}{h^2} \int K'\left(\frac{x-u}{h}\right) f(u) du \\ &= g_h'(x), \end{aligned}$$

using dominated convergence.

Recall that for any $h < \varepsilon$, the number of sign changes of g_h' is at least $2k_f - 1$. Assume now that g_h' has more than $2k_f - 1$ sign changes for some $h < \varepsilon$. Then also $g_{m,h}'$ must have $l > 2k_f - 1$ sign changes for large enough m . (To see this use the convergence of $g_{m,h}'$ to g_h' at $l+1$ points $y_0 < y_1 < \dots < y_l$ chosen such that the points $g_h'(y_i)$ have alternating signs.) We now show that this is impossible: For every m , f_m is almost everywhere differentiable with at most $2k_f - 1$ sign changes in the derivative. (If f is increasing/decreasing on I so is f_m .) Since f_m is Lipschitz continuous and a.e. differentiable, we may use dominated convergence to obtain

$$g_{m,h}'(x) = \frac{d}{dx} \int K(s) f_m(x - hs) ds = \frac{1}{h} \int K\left(\frac{x-u}{h}\right) f_m'(u) du.$$

Assertion (P1) now follows by applying Theorem 3.1(b) of Karlin (1968) to $g_{m,h}'$ and f_m' . The theorem states that for strictly sign regular kernels $K(\cdot)$ and some function $g \neq 0$ a.e., the number of zeroes of $\int K(x-y)g(y)dy$ does not exceed the number of sign changes of g . Notice that strictly totally positive kernels like for instance the Gaussian ($N(0, h^2)$) kernel with arbitrary bandwidth h are strictly sign regular. Notice that our arguments imply that (P1) holds for any $h < \varepsilon$.

Next we show that (P2) holds for any $h < \varepsilon$; i.e. if $g_h'(x) = 0$, then g_h' changes sign at x . Observe first that for any bandwidths $h_1 > 0$ and $h_2 > 0$ satisfying $h = \sqrt{h_1^2 + h_2^2}$ we have that

$$g_h'(x) = g_{\sqrt{h_1^2 + h_2^2}}'(x) = \frac{1}{h_2} \int K\left(\frac{x-u}{h_2}\right) g_{h_1}'(u) du, \quad (5)$$

when $K(\cdot)$ is the Gaussian kernel. This is most easily seen by observing that g_{h_1} is the density of the sum of independent random variables $X + Y_1$, where X has density f and $Y_1 \sim N(0, h_1^2)$. Take now $Y_2 \sim N(0, h_2^2)$. Obviously the density of $(X + Y_1) + Y_2$ equals that of $X + (Y_1 + Y_2)$, with the density of $Y_1 + Y_2$ being equal to the Gaussian kernel with bandwidth h . Eq. (5) follows by taking derivatives.

Now according to (P1), g'_{h^*} has $2k_f - 1$ sign changes for all $h^* < \varepsilon$ and thus in particular for g'_h and g'_{h_1} both have the same number of sign changes. From this (P2) follows by writing $g'_h(x)$ as in (5) and applying again Theorem 3.1(b) of Karlin (1968).

We finally show (P3). We do this by showing that in this case there is a small enough bandwidth $h_* > 0$ and an open neighborhood U_x for each point $x \in A$, such that $\sqrt{h^2 + h_*^2} < \varepsilon$ and furthermore both $g'_{\sqrt{h^2 + h_*^2}}(y) = 0$ for exactly one $y \in U_x$ and $g''_{\sqrt{h^2 + h_*^2}}(z) \neq 0$ for all $z \in U_x$.

Notice that again due to Theorem 3.1(b) of Karlin (1968), the situation $g'_{\sqrt{h^2 + h_*^2}}(y) = 0$ cannot occur, if y is not contained in any of the neighborhoods U_x , $x \in A$.

Observe also that (no matter what h is chosen) it is impossible that both g'_h and g''_h change sign at the same point x . (If g''_h changes sign at x , then g'_h has a local extremum at x .) Therefore there is an open neighborhood $U_x^{(1)}$ for each $x \in A$ such that either $g''_h(z) < 0$ or $g''_h(z) > 0$ for all $z \in U_x^{(1)}$, $z \neq x$. Now for each $x \in A$ there is another open neighborhood $U_x \subseteq U_x^{(1)}$ of x and a small enough h_* such that (for $z \in U_x$) the function $g''_{\sqrt{h^2 + h_*^2}}(z)$ can (up to a negligible part) be written as the integral of an either positive or negative function.

Thus either $g''_{\sqrt{h^2 + h_*^2}}(z) < 0$ or $g''_{\sqrt{h^2 + h_*^2}}(z) > 0$ on U_x . This implies that $g'_{\sqrt{h^2 + h_*^2}}$ is strictly monotonous on U_x and thus there is exactly one zero. It is easily verified that for small enough h_* this zero will be within U_x . $\square$

We next obtain a lower bound on $H_{n,k}$, indicating how small bandwidths can possibly be when ensuring a desired number of modes. Notice that (A1) and (A3) in particular imply that f is of bounded variation.

Lemma 2. Let $h_n = cn^{-\alpha}$ for some arbitrary $c > 0$ and $\alpha \geq \frac{1}{3}$. Then for any probability density f of bounded variation and any finite bound k on the number of modes

$$P[H_{n,k} \geq h_n] \rightarrow 1.$$

Proof. The idea behind the proof is that the variance of $\hat{f}'_{n,h_n}$ does not tend to zero for $n \rightarrow \infty$ if h_n is chosen as required. We will take an arbitrarily large l and show that the probability that the number of sign changes of $\hat{f}'_{n,h_n}$ is at least l tends to one.

With later specified m , take a set S of $2m$ points $x_1 < x_2 < \dots < x_{2m}$ at which f is differentiable and not equal to zero. This is possible since functions of bounded variation are almost everywhere differentiable. Thus, for an x at which f is differentiable,

$$\begin{aligned} E\hat{f}'_{n,h_n}(x) &= \frac{1}{h_n} \int K'(v)f(x - h_nv) dv = - \int vK'(v) \left(\frac{f(x - h_nv) - f(x)}{-h_nv} \right) dv \\ &= - \int vK'(v)(f'(x) + o(1)) dv \rightarrow f'(x), \end{aligned}$$

and there is a positive constant c_1 such that $|E\hat{f}'_{n,h_n}(x)| < c_1$ for all $x \in S$. Furthermore there is some constant c_2 such that $\text{Var}[\hat{f}'_{n,h_n}(x)] > c_2/nh_n^3 \geq c_2/c^3 =: c_4 > 0$. According to standard arguments, the vector

$$\left([\hat{f}'_{n,h_n}(x) - E\hat{f}'_{n,h_n}(x)] / \sqrt{\text{Var}(\hat{f}'_{n,h_n}(x))} \right)_{x \in S}$$

converges in distribution to a vector of independent normal $N(0, 1)$ random variables. Together, we have that for $2\delta = \Phi(-c_1/\sqrt{c_4})$ and for large enough n both $P(\hat{f}'_{n,h_n}(x) > 0) \geq \delta$ and $P(\hat{f}'_{n,h_n}(x) < 0) \geq \delta$ for any $x \in S$.

Partition now S into m subsets $S_i = \{x_{2(i-1)+1}, x_{2(i-1)+2}\}$, $1 \leq i \leq m$. Thus the probability that $\hat{f}'_{n,h_n}(y)$ alternates in sign on S_i is at least $2\delta^2$ for all S_i and large enough n . Now for an arbitrarily small $\varepsilon > 0$ and large enough n , the probability that $\hat{f}'_{n,h_n}$ has at least l ($\leq m$) sign changes can be bounded from below by the

binomial probability

$$1 - \sum_{i=0}^{l-1} \binom{m}{i} (2\delta^2)^i (1 - 2\delta^2)^{m-i} - \varepsilon \geq 1 - r^{m-l} m^l - \varepsilon$$

for $r = 1 - 2\delta^2$ which can be made arbitrarily close to 1, by choosing m and n large enough. $\square$

3. Consistency under modality constraints

We show that any bandwidth selection method will provide consistent (at continuity points) estimates satisfying our modality constraints, if the bandwidth is selected by any rule that is constrained to take the bandwidth from a subset of S_n (or $S_n^{(U)}$) chosen not to include too large bandwidths.

Consider for this purpose an arbitrary sequence $\tilde{H}_n$ of upper bounds, satisfying $\tilde{H}_n \rightarrow 0$ a.s. and define

$$\tilde{S}_n = \{H_{n,k}\} \cup [H_{n,k}, \min[\max(H_{n,k}, \tilde{H}_n), H_{n,k-1}]].$$

A possible choice for $\tilde{H}_n$ would be $\tilde{H}_n := n^{-\gamma}$ for some positive $\gamma < \frac{1}{3}$. In the case where only an upper bound on the number of modes is specified, we proceed analogously and define

$$\tilde{S}_n^{(U)} = [H_{n,k}, \max(H_{n,k}, \tilde{H}_n)].$$

Notice that $\tilde{S}_n$ and $\tilde{S}_n^{(U)}$ are always nonempty.

The following theorem establishes consistency for any bandwidth selection method that chooses a bandwidth from the set $\tilde{S}_n^{(U)}$. Since $\tilde{S}_n \subseteq \tilde{S}_n^{(U)}$, the result holds also for $\tilde{S}_n$.

Theorem 1. Suppose that (A1)–(A3) hold and that $k \geq k_f$. Then as $n \rightarrow \infty$

$$\sup_{h \in \tilde{S}_n^{(U)}} |\hat{f}_{n,h}(x) - f(x)| \rightarrow 0 \quad \text{in prob.}$$

if f is continuous at x .

Proof. Proceed as in (3) to show that

$$\sup_{h \in \tilde{S}_n^{(U)}} \sup_x |\hat{f}_{n,h}(x) - \mathbb{E} \hat{f}_{n,h}(x)| \leq C \|F_n - F\|_\infty \sup_{h \in \tilde{S}_n^{(U)}} \frac{1}{h},$$

which tends to zero in prob., since $H_{n,k} \geq n^{-1/3}$ in prob. according to Lemma 2, and $P(\|F_n - F\|_\infty \geq h\varepsilon) \leq c \exp(-\varepsilon^2 nh^2)$ for any $\varepsilon > 0$ according to the Dvoretzky–Kiefer–Wolfowitz inequality. See for instance [Shorack and Wellner \(1986\)](#).

For the bias we have that

$$\sup_{h \in \tilde{S}_n^{(U)}} |\mathbb{E} \hat{f}_{n,h}(x) - f(x)| \leq \sup_{h \in \tilde{S}_n^{(U)}} \int K(u) |f(x - hu) - f(x)| du,$$

which tends to zero a.s. for $n \rightarrow \infty$, since $H_{n,k} \rightarrow 0$ a.s. according to Lemma 1. $\square$

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